

KRUSHI RAJ TULA

Senior Full Stack Engineer | Search & Data Platforms, AI Agent Systems
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SUMMARY

Senior full-stack engineer with **7+ years** building search and data platforms at scale, and the AI tooling that runs over them. At Paperpile I am one of a handful of engineers and the only one on the academic search platform. I came in on crawlers and ingestion, took over author parsing and search performance, and now own it end to end: **230M+ documents, 300GB+ of text a month**, crawlers across **500+ publisher domains**. I set technical direction, lead cross-functional projects, mentor, and carry the pager for what I ship.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Node.js, Python, Go, Rust

AI & Agents: MCP server development, agent harnesses, evaluations and scoring gates, multi-agent orchestration, LLM integration, hallucination mitigation, Claude Code, Cursor

Data & Search: MongoDB, Elasticsearch, PostgreSQL, Redis, ClickHouse, SQLite, indexing, query optimization, large-scale ingestion, vector and semantic search

Backend: Microservices, REST, GraphQL, RPC pipelines, RabbitMQ, WebSockets, event sourcing, idempotency, fault tolerance

Frontend: React, Next.js, Tailwind CSS, internal dashboards and data tooling

Cloud & Ops: Docker, Kubernetes, Jenkins, CI/CD, AWS S3, AWS Lambda, ELK, ClickHouse telemetry, alerting, on-call

WORK EXPERIENCE

Senior Full Stack Engineer, Paperpile, Remote · Jan 2020 - Present

- Started on **crawlers and ingestion** and still own them: scraper engines across 500+ publisher domains, 300GB+ of text monthly.
- Built the author-name parser to **87% accuracy against a 45-58% baseline**, and raised abstract coverage of the corpus from 27% to 51%.
- Reduced **p99 search latency from ~10s to under 1s** by profiling the request path and redesigning indexing and query strategy.
- Now own the platform end to end at 230M+ documents**, from architecture and APIs to deployment, monitoring and backups, and set what gets built next.
- Lead cross-functional projects:** coordinated the citation-verification release to its planned date, and led crawler development, parsers and imports, developer experience, Microsoft login and OneDrive sync.
- Managed a master's-level intern** for a 2-3 month project on journal and database suggestions, the team's first LLM-based work.
- Shipped an **MCP server** so LLM tools verify citations against the corpus, and a test-gated agent harness that **cut a release cycle from ~2 weeks to ~2 days**.
- Designed **10+ microservices** passing millions of messages with strict retries, ordering and idempotency.
- Cut production error rate from 10% to 2%** and deploy time by half on Docker, Kubernetes and Jenkins; run observability on ELK and ClickHouse.
- Found and fixed a reporting defect that had silently hidden **5.67M documents** from stakeholder-facing numbers. Nobody assigned it.
- Work directly with support to triage user reports, separate real defects from noise, and fix them.

Environment: TypeScript, Node.js, Python, MongoDB, Elasticsearch, PostgreSQL, ClickHouse, Redis, RabbitMQ, AWS S3, Docker, Kubernetes, Jenkins, ELK, MCP.

Founding Engineer (contract engagement), early-stage fintech startup, Remote · 2023

- Hired the first engineering team from zero:** defined the JD, screened and interviewed with the founders, and scored candidates so the founders could set pay. **3 engineers in about 2 weeks**.
- Chose **Go** as the founding stack after analysing the stacks and operations of comparable fintech companies, on the grounds that the domain needed determinism.

Freelance Engineer & Team Lead · 2021 - 2026

- Built a React Native app and **led 4 developers** delivering the companion web app, coordinating the work and bringing them up to speed on WooCommerce.
- For dSynths, a DeFi trading platform, **cut average chart render from ~1s to ~10ms** with Redis caching; separately wrote and deployed a Solidity contract with an ETH and ERC-20 mint flow.

Software Engineer, NCR Corporation, Hyderabad, India · Jul 2019 - Dec 2019

- Cut ATM ad load latency from ~2 minutes to under 30 seconds** on low-memory, low-bandwidth hardware, and unblocked a major bank chain's campaign by root-causing the defect stopping it.

Software Developer Intern, ServiceNow, Hyderabad, India · Jan 2019 - Jun 2019

- Contributed to a proprietary internal frontend framework and to a patented activity-monitoring system.

PROJECTS

- **posix.co.in** (2024-present): production SaaS for retail and pharmacy sellers, **live with 4 partner businesses** I onboarded myself, with telemetry driving what got fixed first.
- **Royal Lands** (2026): multiplayer game with a server-authoritative Go backend, one goroutine owning each room's state, and **event-sourced replay so games survive redeploys**. royal-lands.up.railway.app
- **Multi-agent content harness** (2026): sub-agents draft and score independently until a version clears a quality gate. **Cut production time ~10x**. instagram.com/thetechniches
- **code-threat-scan** (2026): open-source scanner for planted malware in untrusted repos, written after two real cases. github.com/krushiraj/code-threat-scan
- **Open source**: merged PRs to fetch-cookie and mislav/gh-contrib, proposed fixes to PocketBase, three self-authored Homebrew taps. github.com/krushiraj

TEACHING

Adjunct Faculty / Guest Lecturer, Vasavi College of Engineering, Hyderabad · 2022 - Present

- Teach full-stack development and DevOps, mentor engineers through project reviews and mock interviews, and judged a national-level hackathon.

EDUCATION

Bachelor of Engineering, Information Technology, Vasavi College of Engineering · 2015 - 2019